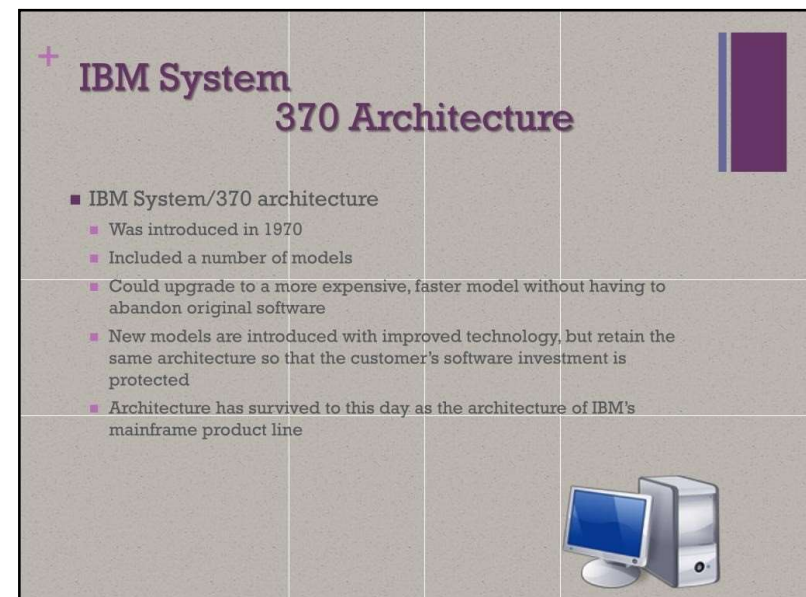
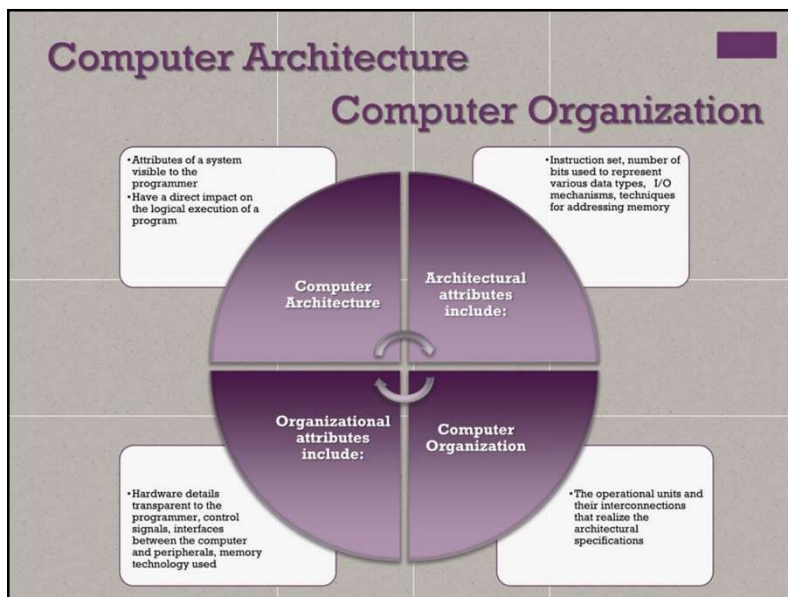
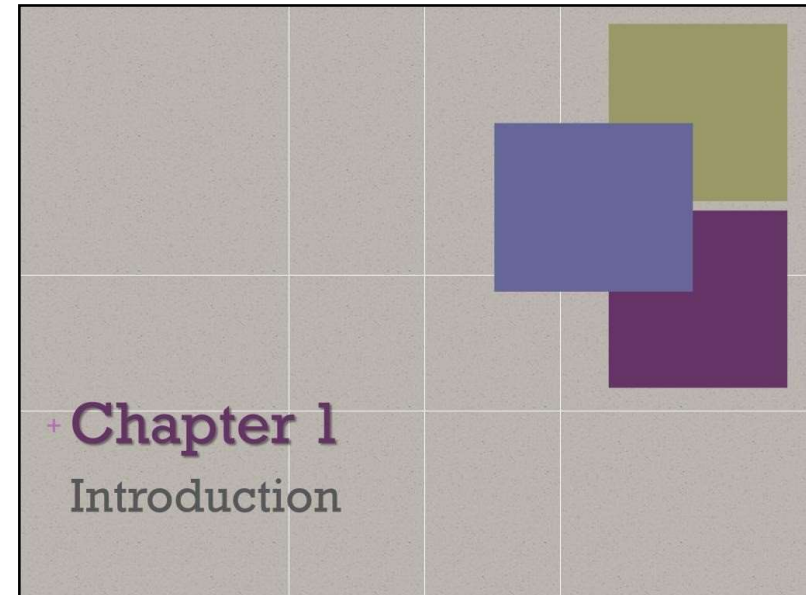
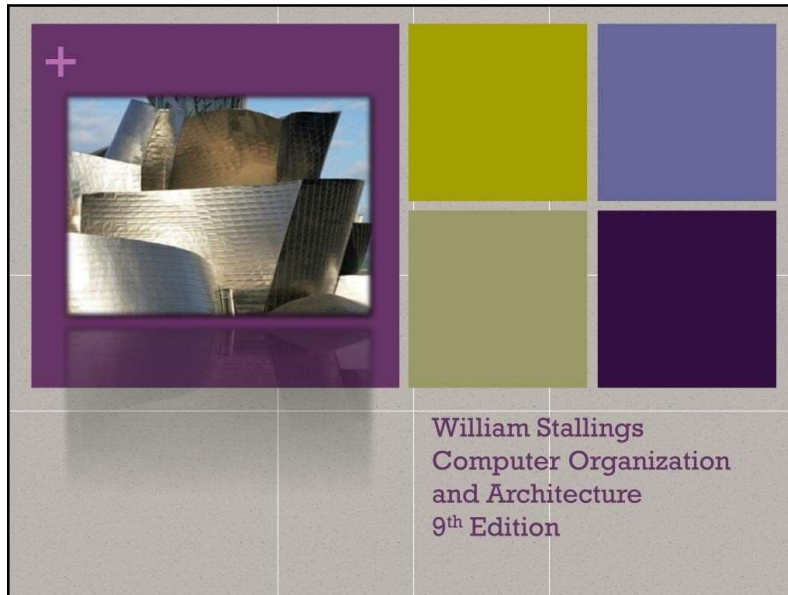




+ Structure and Function

- Hierarchical system
 - Set of interrelated subsystems
- Hierarchical nature of complex systems is essential to both their design and their description
- Designer need only deal with a particular level of the system at a time
 - Concerned with structure and function at each level
- Structure
 - The way in which components relate to each other
- Function
 - The operation of individual components as part of the structure



Function

A computer can perform four basic functions:

- Data processing
- Data storage
- Data movement
- Control

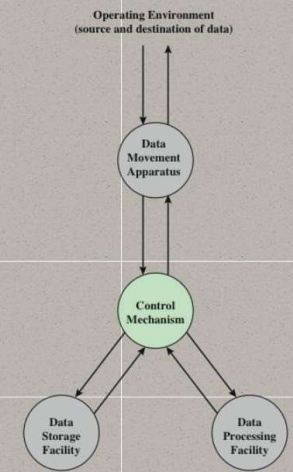


Figure 1.1 A Functional View of the Computer



Operations

(a)
Data movement

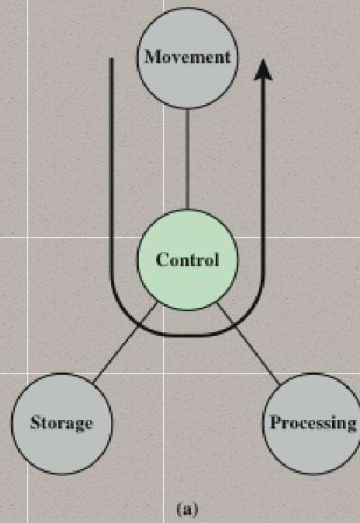


Figure 1.2 Possible Computer Operations



Operations

(b)
Data storage

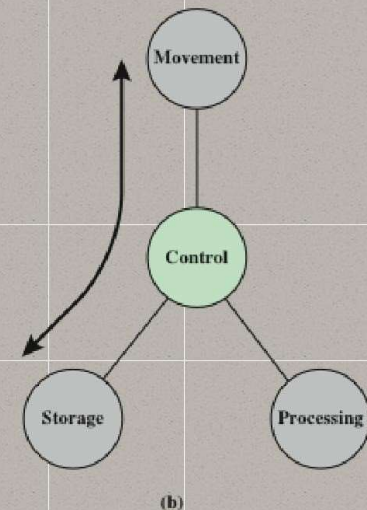
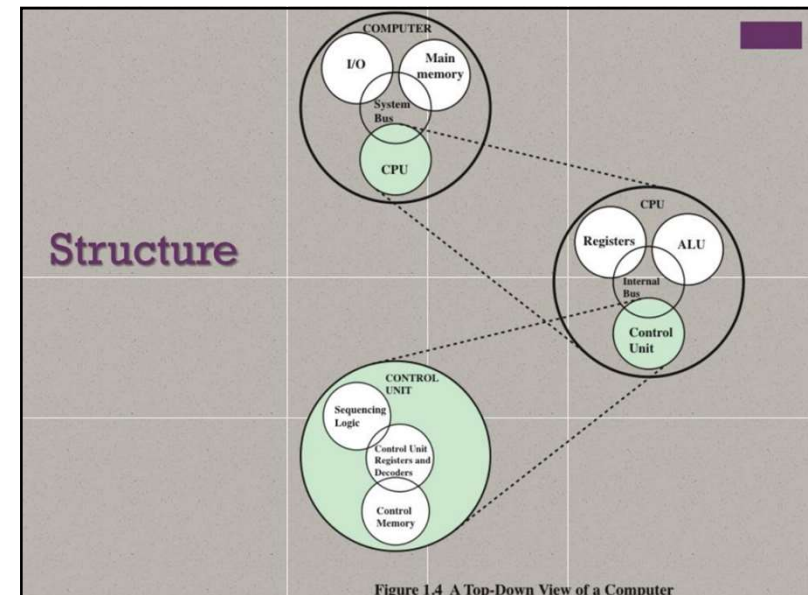
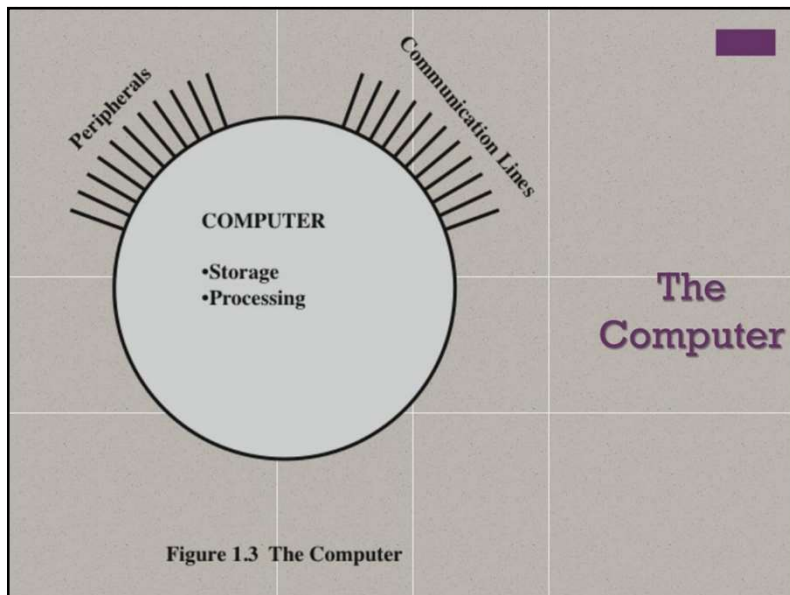
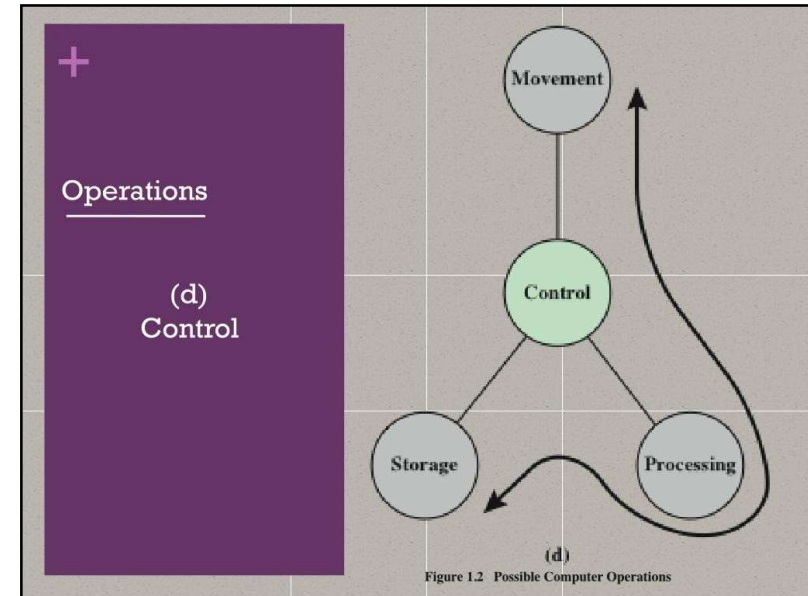
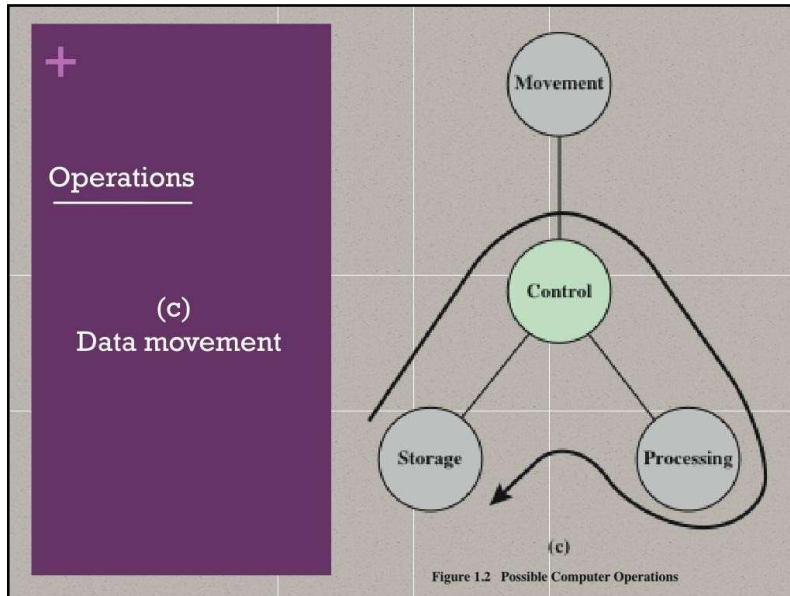


Figure 1.2 Possible Computer Operations





There are four main structural components of the computer:



CPU - controls the operation of the computer and performs its data processing functions

Main Memory - stores data

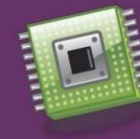
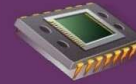
I/O - moves data between the computer and its external environment

System Interconnection - some mechanism that provides for communication among CPU, main memory, and I/O



CPU

Major structural components:



- Control Unit
Controls the operation of the CPU and hence the computer
- Arithmetic and Logic Unit (ALU)
Performs the computer's data processing function
- Registers
Provide storage internal to the CPU
- CPU Interconnection
Some mechanism that provides for communication among the control unit, ALU, and registers

+ Summary

Chapter 1

- Computer Organization
- Computer Architecture
- Function
 - Data processing
 - Data storage
 - Data movement
 - Control

Introduction

- Structure
 - CPU
 - Main memory
 - I/O
 - System interconnection
- CPU structural components
 - Control unit
 - ALU
 - Registers
 - CPU interconnection

+ Internet Resources - Web site for book

- <http://WilliamStallings.com/COA/COA9e.html>
 - Links to sites of interest
 - Links to sites for courses that use the book
 - Errata list for book
 - Information on other books by W. Stallings
- <http://WilliamStallings.com/StudentSupport.html>
 - Math
 - How-to
 - Research resources
 - Misc

ACTIVITY 2

Answer the following questions in a whole sheet of paper.

Based on our discussion:

1. What, in general terms, is the distinction between computer organization and computer architecture?
2. What, in general terms, is the distinction between computer structure and computer function?
3. What are the four main functions of a computer?
4. List and briefly define the main structural components of a computer.
5. List and briefly define the main structural components of a processor.